

Leader Election 2

Chapter 3

Election in the Ring: Stage Technique

Leader Election on Ring: The *Stage* Technique

- **General** Technique:

Protocol works in *Stages*.

Every node knows the Stage he is performing.

Protocol Analysis: at the end of every Stage,
Some **global property** holds.

Controlled Distance

Basic idea: Operate in stages. An entity maintains control on its own message.

ASSUMPTIONS

- Bidirectional ring
- Different *ids*
- Local orientation

sense of direction only for simplicity - not needed

Ingredients

1) Limited distance (to avoid big msgs to travel too much)

Ex: stage i : distance 2^{i-1}

2) Return messages (if seen something smaller does not continue)

3) Check both sides

4) Smallest always win (regardless of stage number)

Candidate entities begin the algorithm.

Stage i:

- Each *candidate* entity sends a message with its own **Id** in both directions
- the msg will travel until it encounters a smaller Id or reaches a certain distance (2^{i-1})
- If a msg does not encounter a smaller Id, it will return back to the originator (feedback msg)



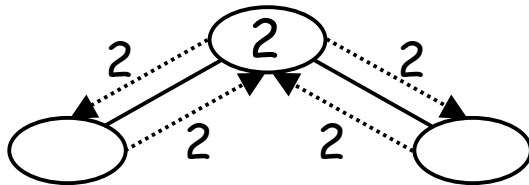
- A candidate receiving its own feedback-msg back from both directions survives and start the next stage

Entities encountered along the path read the message and:

- Each entity i with a **greater** identity Id_i becomes **defeated** (passive).
- A **defeated** entity forwards the messages originating from other entities, if the message is a **notification** of termination, it terminates

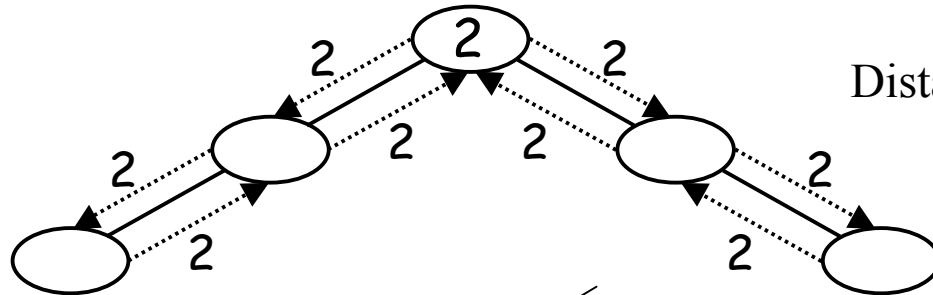
More...

Stage 1



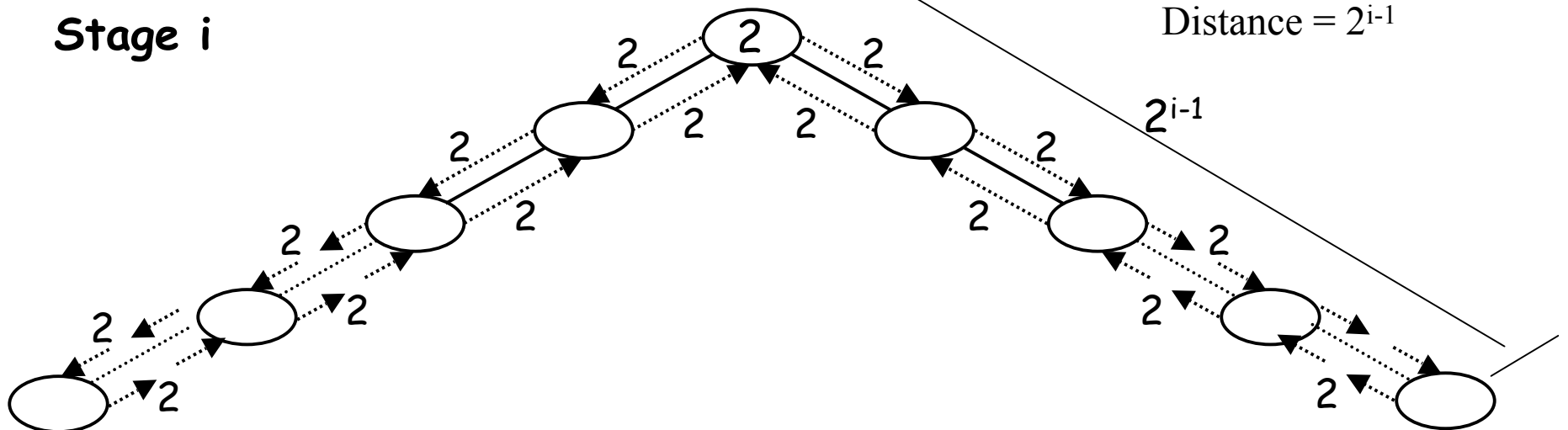
Distance = 1

Stage 2



Distance = 2

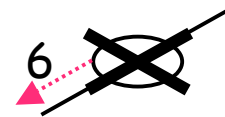
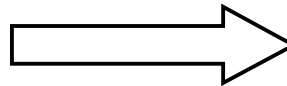
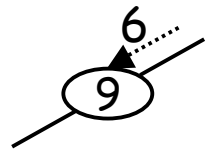
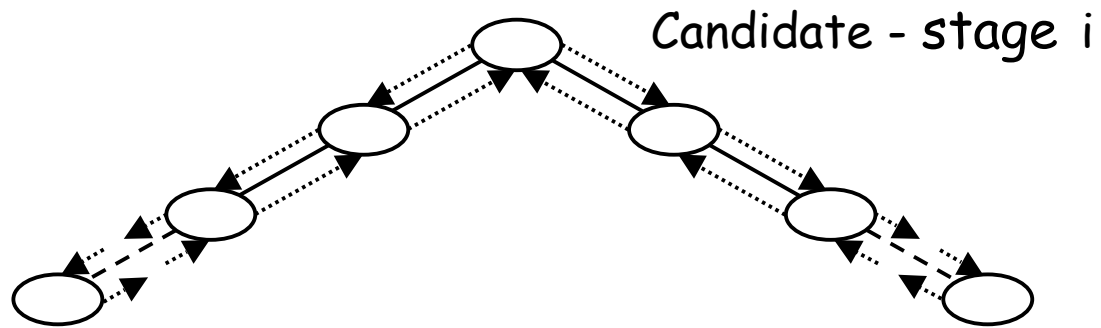
Stage i



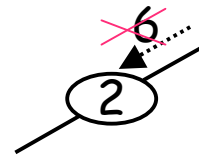
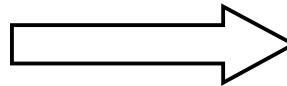
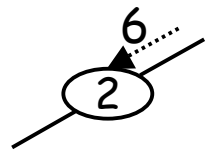
Distance = 2^{i-1}

2^{i-1}

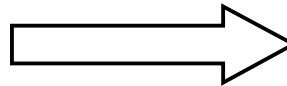
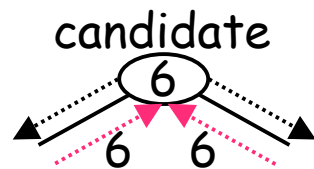
More...



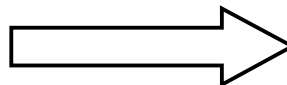
node is defeated



msg is defeated



stage i + 1



Becomes passive



Correctness and Termination

If a candidate receives one (left/right) of its msgs from the opposite side (right/left) it sent it, it becomes the leader and notifies.

- The smallest id will always travel the max distance defeating every entity it encounters
- The distance monotonically increases eventually becoming greater than n
- The leader will eventually receive its message from the opposite directions

Notes:

- we do not need message ordering.

-What happens if an entity receives a message from a higher stage ?

Possible *bad* situation: a node is defeated by some msg of stage 8 and, after that, he receives his feedback msg of stage 7.

SOLUTION: GREEDY ALWAYS WORKS !!!

If at any time a node receives a smaller id (of any stage), it will become defeated and will not wait anymore for his feedback msgs

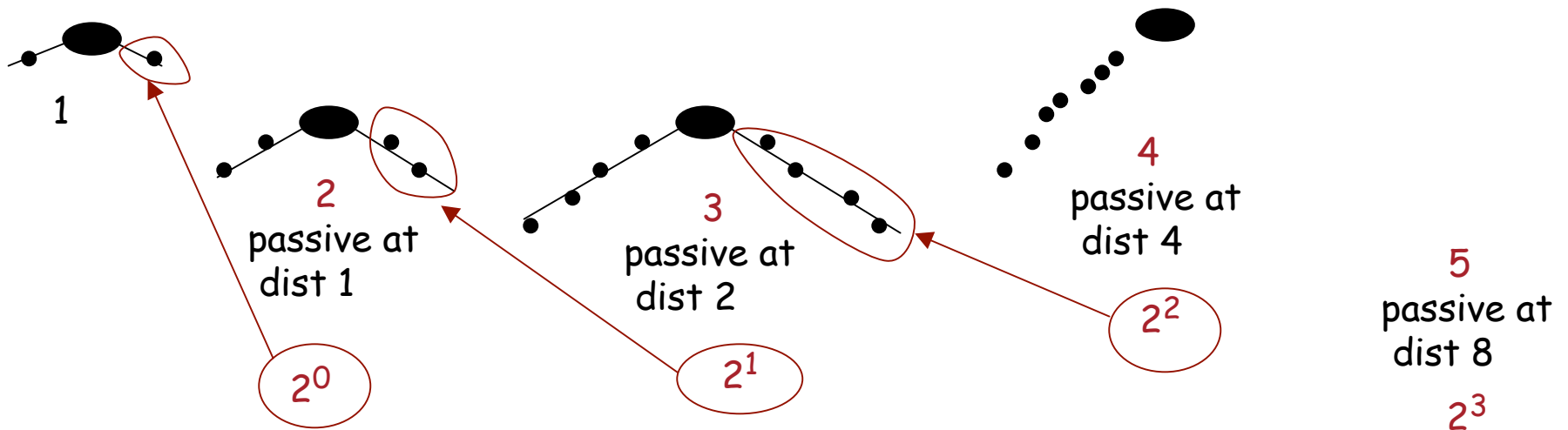
Message Complexity

When the distance is doubled at each stage
i.e., $\text{dis}(i) = 2^{i-1}$:

Notion of Logical Stage

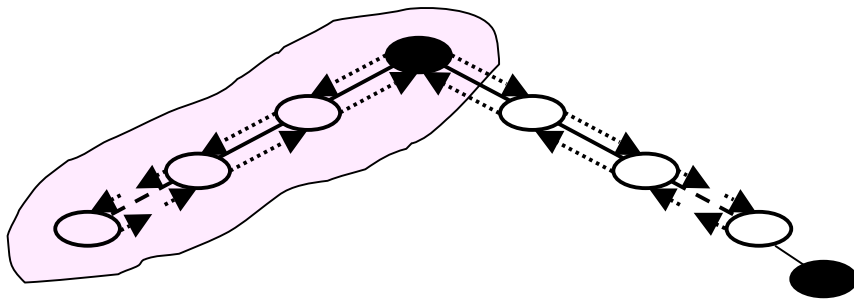
Def: n_i entities start stage i

If x starts stage i (i.e., survived stage $i-1$) the Id of x must be smaller than the Ids of the neighbours at distance up to 2^{i-2} on each side



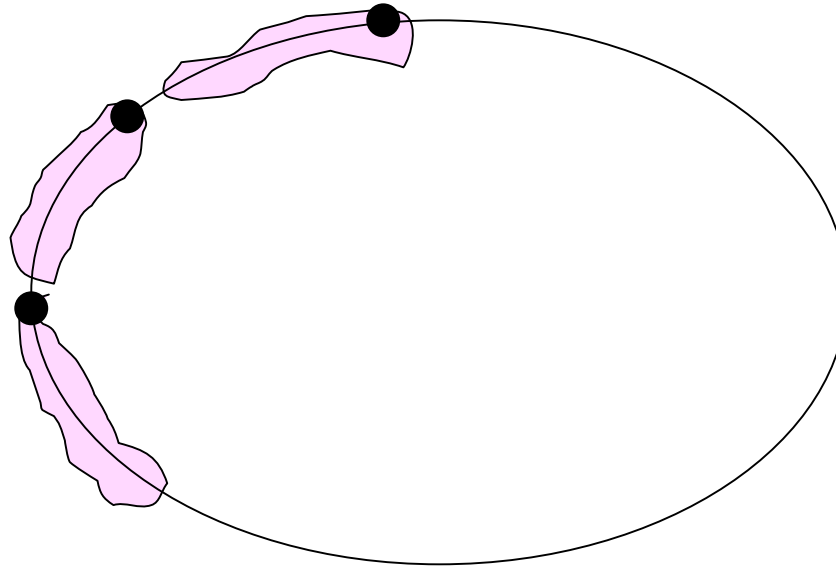


Within any group of $2^{i-2} + 1$ consecutive entities
at most one can survive stage $i-1$.



So,

$$n_i \leq n / (2^{i-2} + 1)$$



Starting stage i :

"Forth" messages:

each will travel at most 2^{i-1} in both directions

$$\text{Tot: } 2 n_i 2^{i-1}$$

"FeedBack" messages:

each survivor will receive one from each side

$$2 n_{i+1} 2^{i-1}$$

each entity that started the stage but did not survive
will receive none or one, so :

$$\leq (n_i - n_{i+1}) 2^{i-1}$$

$$\text{Tot: } 2 n_{i+1} 2^{i-1} + (n_{i+1} - n_i) 2^{i-1}$$

stage $i > 1$

$$\text{Tot: } 2 n_i 2^{i-1} + 2 n_{i+1} 2^{i-1} + (n_{i+1} - n_i) 2^{i-1}$$

$$= (3 n_i + n_{i+1}) 2^{i-1}$$

$$n_i \leq n / (2^{i-2} + 1)$$

$$\leq (3 \lfloor n / (2^{i-2} + 1) \rfloor + \lfloor n / (2^{i-1} + 1) \rfloor) 2^{i-1}$$

$$< \frac{3 n 2^{i-1}}{(2^{i-2} + 1)} + \frac{n 2^{i-1}}{(2^{i-1} + 1)}$$

$$< \frac{3 n 2^{i-1}}{2^{i-2}} + \frac{n 2^{i-1}}{2^{i-1}}$$

$$= 6 n + n = 7 n$$

The first stage is a bit different:

If everybody starts:

the survivors $4 n_2 2^0$ 2 "forth", 2 "back"

the others $3 (n - n_2) 2^0$
2 "forth", 1 "back"

$$n_2 \leq n / (2^0 + 1)$$

$$4 n_2 + 3 n - 3 n_2 = n_2 + 3 n$$

$$< 4n$$

Total Number of Stages

The ring is fully traversed as soon as 2^{i-1} is greater than or equal to n

$$2^{i-1} \geq n$$

That is, when:

$$i \geq \log n + 1$$

-----> $\log n + 1$ stages

first stage

$$TOT \leq \sum_{i=1}^{\log n} 7n + O(n)$$

$$= n \sum_{i=2}^{\log n} 7 = 7 n \log n + O(n)$$

$$O(n \log n)$$

Conjecture:

In unidirectional rings,
the worst case complexity is (n^2) ;
to have a complexity of $O(n \log n)$ messages,
bidirectionality is necessary.

NOT TRUE !

Stages

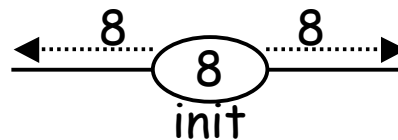
Basic idea:

A message will travel until it reaches another candidate
A candidate will receive a message from both sides

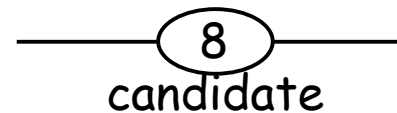
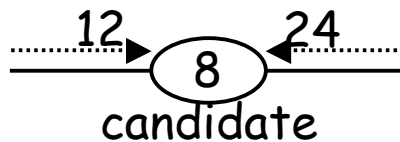
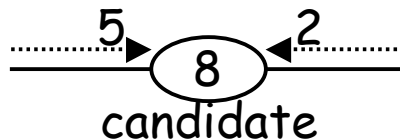
ASSUMPTIONS

- Distinct *ids*
- Bidirectional ring (+ unidirectional version)
- Local orientation
- Message ordering (for simplicity only: not needed)

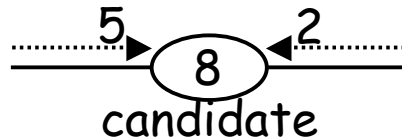
Each *candidate* sends its own Id in both directions.



When a *candidate* i receives two messages Id_j (from the right) and Id_k (from the left), it determines if it becomes *passive* (= it is not the smallest), or if it remains *candidate* (= it is the smallest).



When a *candidate* i receives two messages Id_j (from the right) and Id_k (from the left),



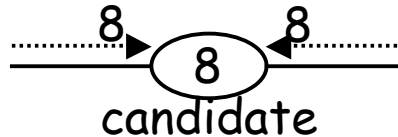
After receiving the first message:
close-port (enqueue messages possibly arriving later)

After receiving the second message, perform the action
and **re-open-port**

Correctness and termination

The minimal entity will never cease to send messages.

When an entity knows that it is the *leader*

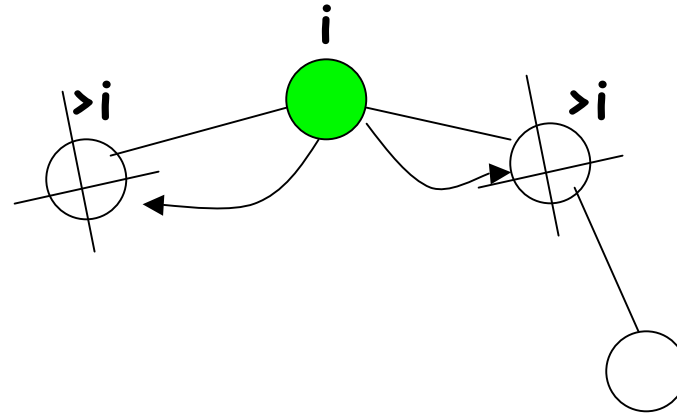


it sends a *notification* message which travels around the ring.

Complexity - Worst Case

At each step: At least half the entities became passive.

$$n_{i+1} \leq \frac{n_i}{2}$$



$$\begin{aligned} n_0 &= n \\ n_1 &\leq n/2 \end{aligned}$$

$$n_i \leq n/2^i$$



$$n/2^k \leq 1$$

when

$$k \geq \log n$$

steps: At most $\lfloor (\log n) \rfloor$

Each entity sends or resends 2 messages.

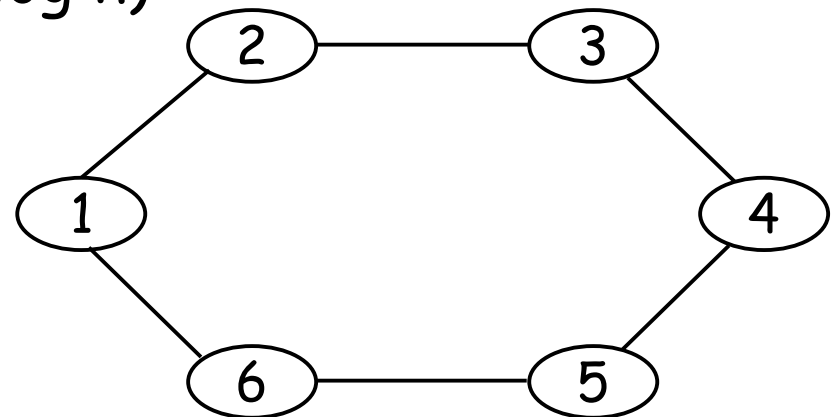
messages: $2n$

bits: $2n * \lfloor (\log n) \rfloor$

Last entity: $2n$ messages to understand that it is the last active entity, then n notification messages.

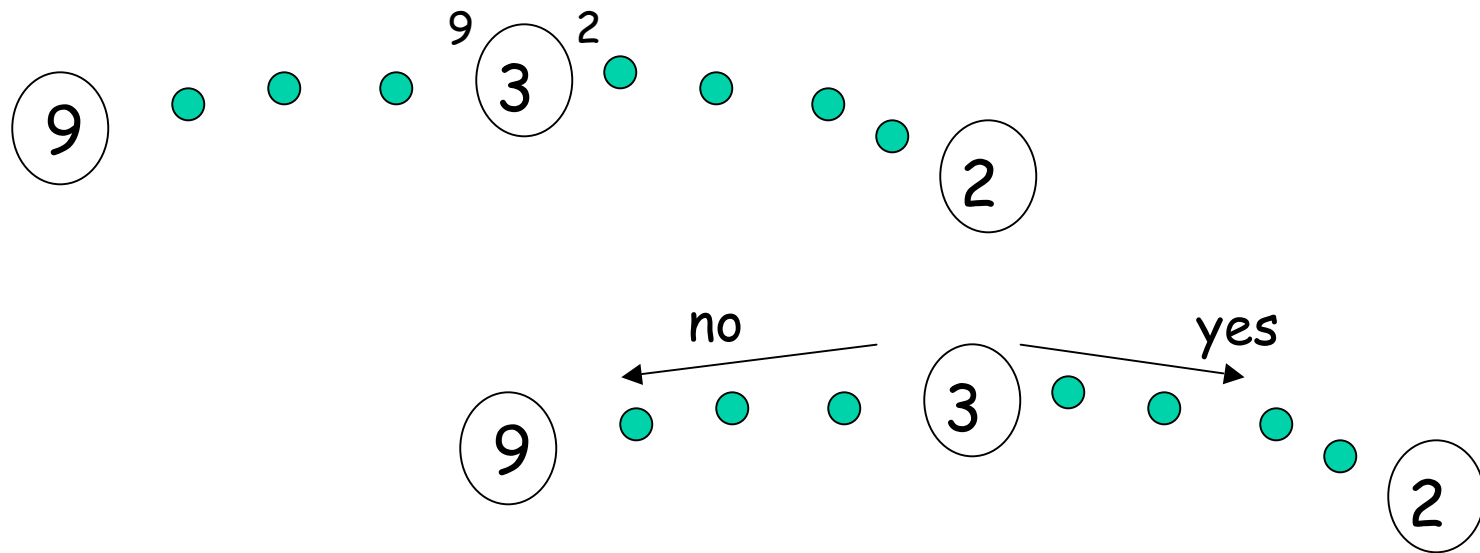
Total: $2n * \lfloor (\log n) \rfloor + 3n = O(n \log n)$

Best Case ?

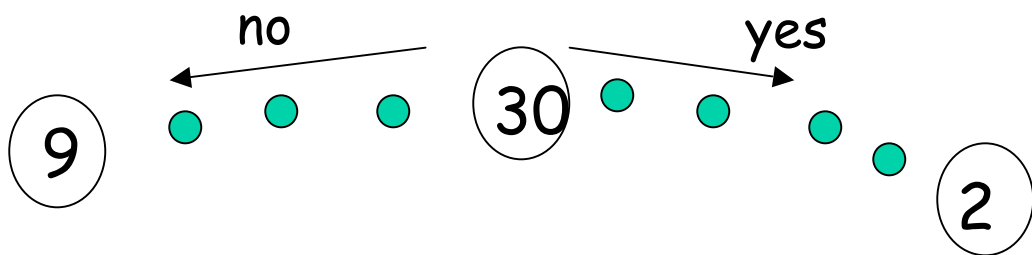
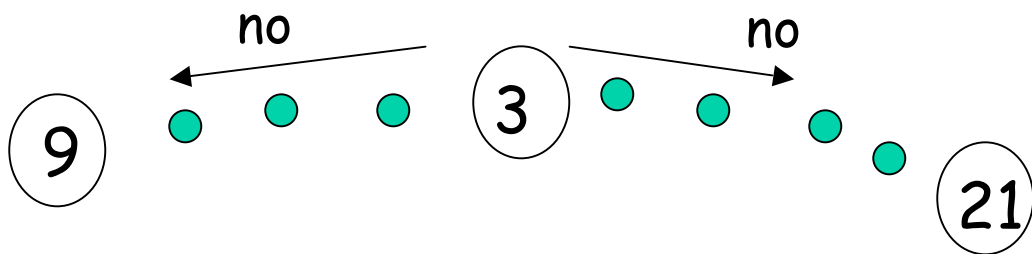


Stages with Feedback

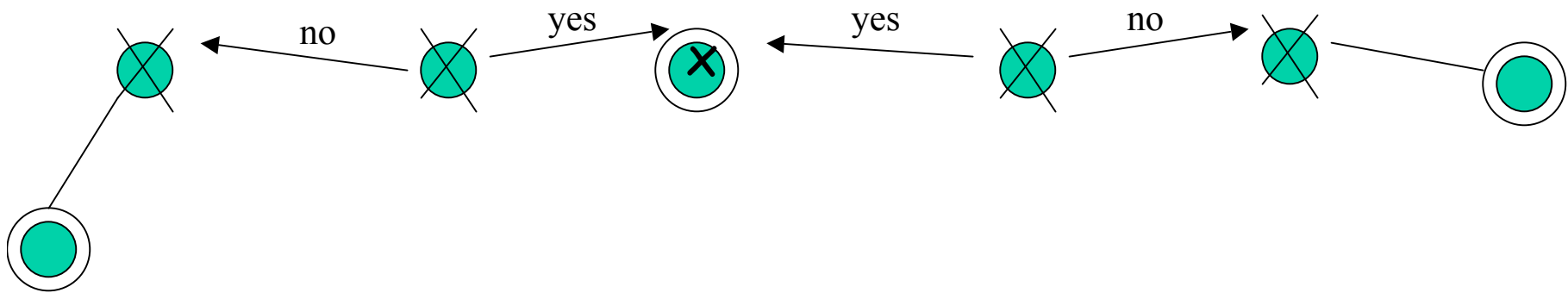
A feedback is sent back to the originator of the message



send YES to the smallest of the two IF it is smaller than me
(otherwise send NO)
send NO to the other



If x survives, it must have received a feedback from both neighbouring candidates ...



$$n_{i+1} \leq \frac{n_i}{3}$$

Unidirectional version

Simulation of the bidirectional algorithm with the same complexity.

Examples

The Conjecture is false.

Alternating Steps

Basic idea: Alternating directions.

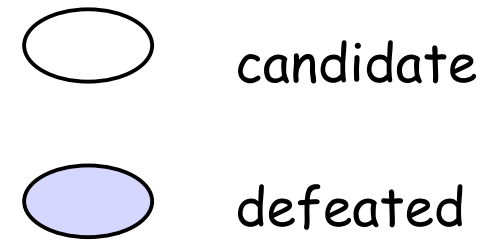
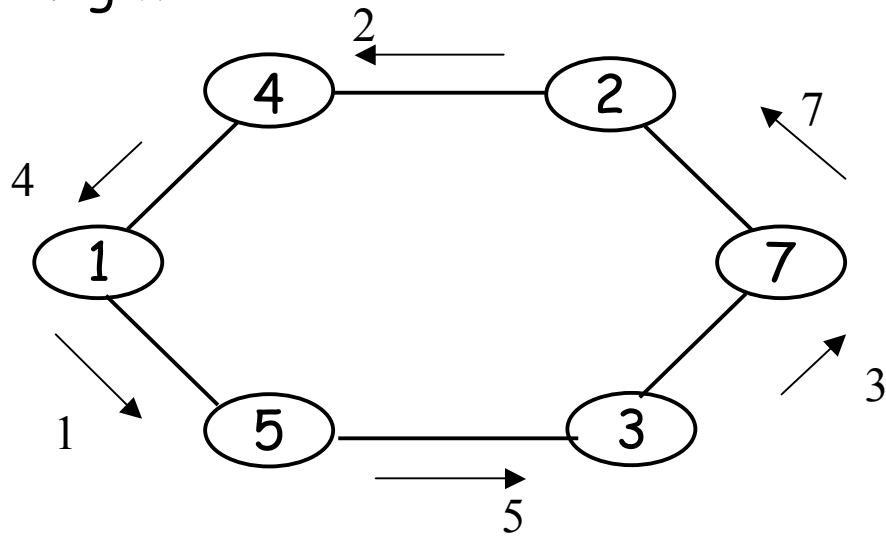
- Different *ids*.
- Bidirectional ring and *sense of direction*.
- Local orientation.
- Message ordering.

send-left
begin-to-defeat (if possible)
send-right

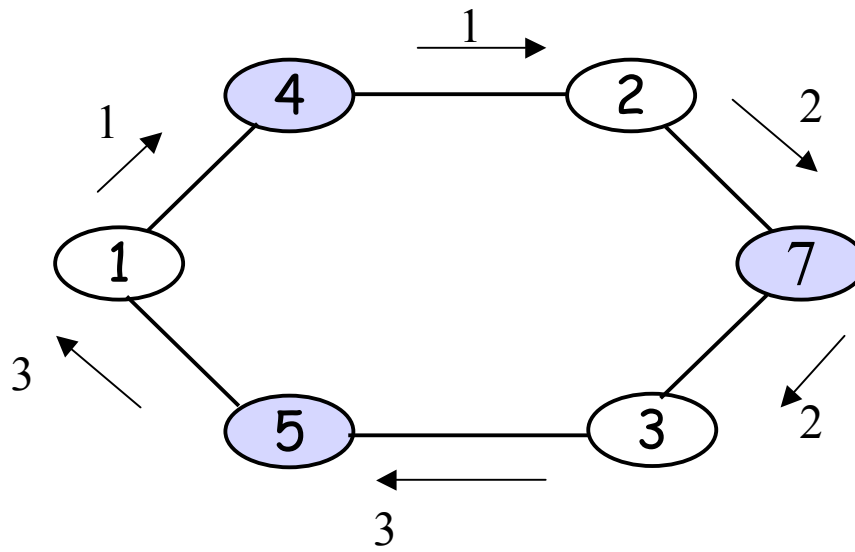
Algorithm:

1. **Each entity** sends a message **to its right**. This message contains the entity's own *id*.
2. Each entity compares the *id* it received from its left to its own *id*.
3. If its own *id* is greater than the received *id*, the entity becomes passive.
4. All entities that remained active (surviving) send their *ids* **to their left**.
5. A surviving entity compares the *id* it received from its right with its own *id*.
6. If its own *id* is greater than the *id* it received, it becomes passive.
7. Go back to step 1 and repeat until an entity receives its own *id* and becomes *leader*.

Step 1: right



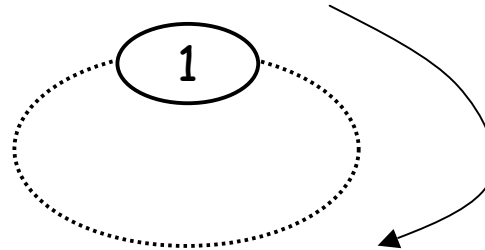
Step 2: left



Complexity

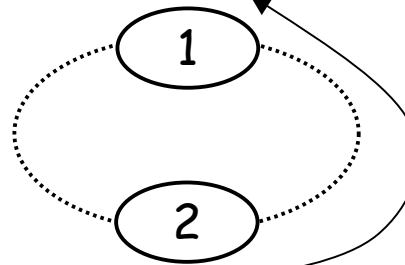
Analyze # of steps in worst case:

Last phase k



1 active entity

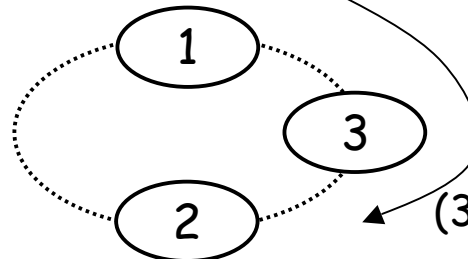
Phase k - 1



at least 2 active entities

(2) will become passive at the next step.

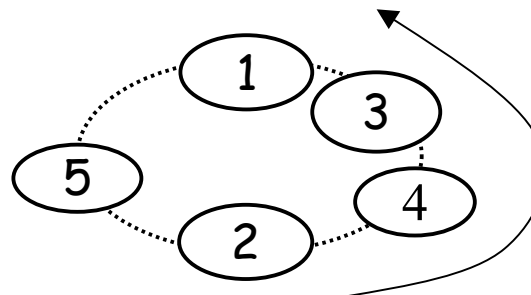
Phase k - 2



at least 3 active entities

(3) must be there; otherwise, (2) would be killed.

Phase k - 2



at least 5 active entities

1 2 3 5 8 13 21

steps =

index of the lowest Fibonacci number $\geq n$

$$F_1 = 1$$

$$F_2 = 2$$

$$F_3 = 3$$

$$F_4 = 5$$

$$F_5 = 8$$

...

$$F_k = i = ?$$

$$= \text{approx. } 1.45 \log_2 n$$

Messages = n for each step

Total = approx. $1.45 n \log_2 n$

upper bounds

Bidirectional		Unidirectional	
LeLann (1977) "All the way"	n^2	LeLann (1977) Unidirectional simulation	n^2
Chang & Roberts (1979) "As far as you can" average case $n \log n$	n^2	Chang & Roberts	n^2
Hirshberg & Sinclair (1980) stages message control	$7n \log n$		
Franklin (1982) stages	$2n \log n$	Dolev, Klawe & Rodeh Unidirectional simulation	$2n \log n$
Peterson (1982) Alternate	$1.44n \log n$	Peterson 1982 Unidirectional simulation	$1.44n \log n$
		Dolev, Klawe & Rodeh (1982)	$1.36n \log n$
		Higham, Przytycka (1984)	$1.22n \log n$

lower bounds

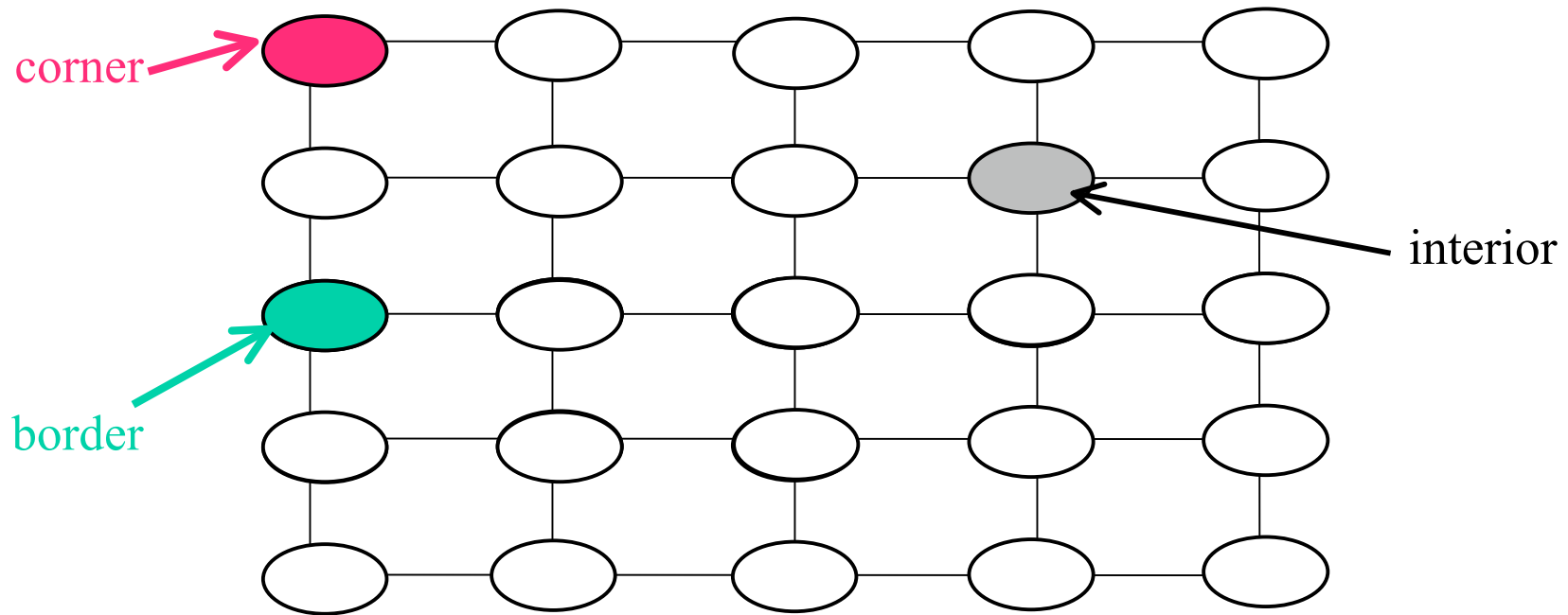
Burns

$$0.5n \log n$$

Pachl, Korach
Rotem (1984)

$$0.69n \log$$

Mesh



$a = \# \text{ rows}$

$b = \# \text{ columns}$

$n = a \times b \text{ nodes}$

$m = a(b-1) + b(a-1) = O(n)$

Important Fact 1: Asymmetric topology

corners

borders

internals

Important Fact 2: The subgraph induced by corners + borders is a **RING** !!!

Exploit that in Election Protocol !!!

Idea: Elect as a **Leader** one of the **four**
corners

Three phases:

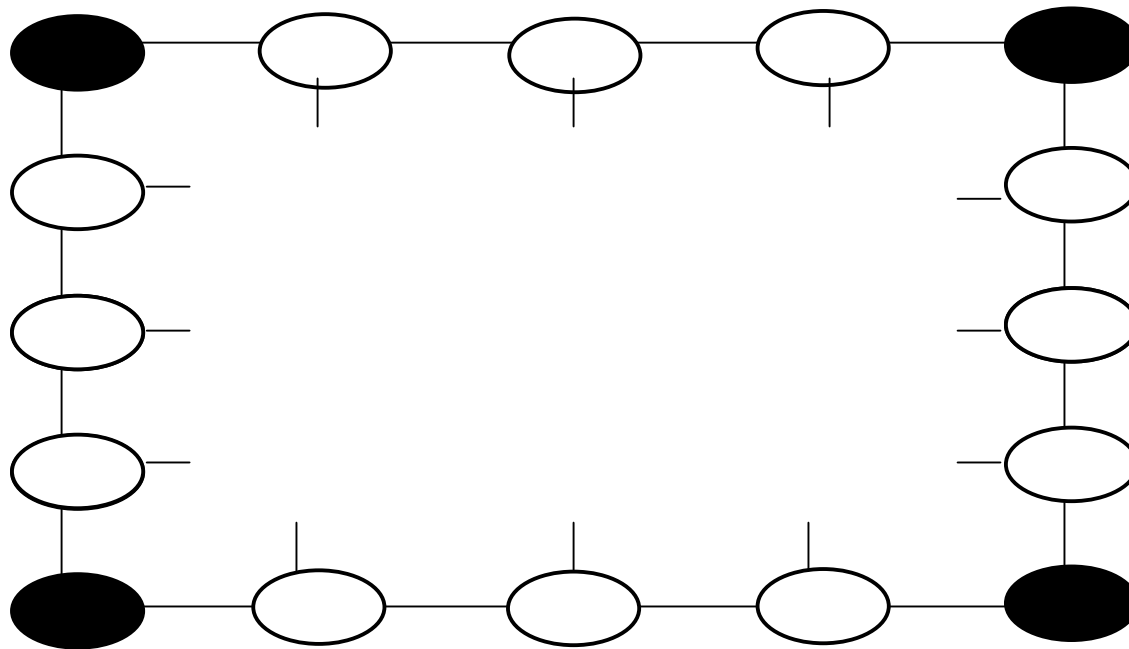
- 1) Wake up
- 2) Election (on the border) only among
corners
- 3) Notification (Broadcast)

1) Wake-up ($k = \#$ initiators)

- Each Initiator sends a **wake-up** to its neighbours
- A non-initiator receiving a **wake-up**, sends it to its other neighbours

$$M(\text{Wake-up}) = 3n + k = O(n)$$

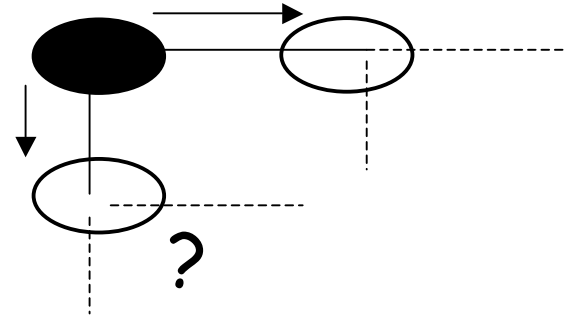
2) Election on the **Ring Border** started by the **Corners**
Corners are the only eligible ones



$$O(\sqrt{n})$$

- Each corner x sends its $\text{id}(x)$ to ALL ($N = 2$)

- Each border forwards every **new** msg to ALL ($N = 3$)



- Each corner forwards every **new** msg to OTHER

- Each interior does NOTHING

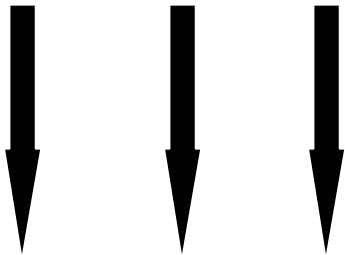
When can nodes **STOP** ??

We are in a very **special** topology!

Crucial FACTS :

Every node knows he is in a MESH!

Every node knows there will be exactly 4 Corner Msgs !



Every “ring” node can stop and decide after he has received 4 Corner Msgs.

MESSAGE COMPLEXITY (ELECTION PHASE)

In every Ring link --> 4 MSGs : $TOT = O(a+b) = O(\sqrt{n})$

In every internal link adjacent to the BORDER:

4 MSGs : $TOT = O(a+b) = O(\sqrt{n})$

$M(\text{Election Phase}) = O(\sqrt{n})$ $o(n) !!!$

NOTIFICATION PHASE

The Corner Leader x BROADCASTS its $Id(x)$ over all MESH.

Unique Initiator

$$M(\text{Flooding} \mid \text{RI}) = O(m) = O(n)$$

Note: The hardest tasks are:

- 1) Wake-up
- 3) Notification

3) Notification

by flooding

$$O(m) = O(n)$$

TOT: $O(n)$

Torus

